Classifier-Free Guidance (CFG)

A Visual Introduction for Beginners

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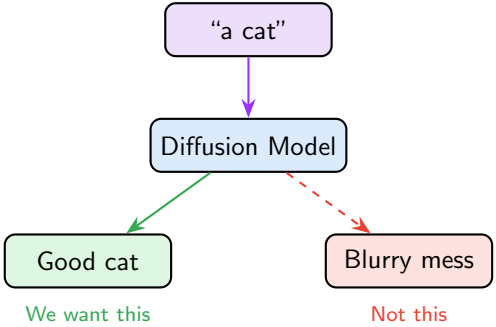
The Problem: Conditional Generation

We want a diffusion model to generate images **matching a text prompt**.

Example:

- Prompt: "a photo of a cat"
- Model should generate: a realistic cat image
- Model should **NOT** generate: random noise, a dog, or a blurry mess

Challenge: Naive conditional models often produce outputs that are *correct but bland* — they lack fidelity and sharpness.



Two Approaches to Guidance

Classifier Guidance

Need a **separate** pretrained classifier

$\nabla_x \log p(c|x)$

(Dhariwal & Nichol, 2021)

$\Longrightarrow$

Can we do better

Classifier-Free Guidance ✓

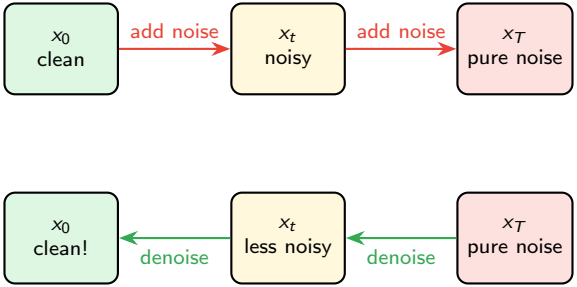
No separate classifier!
Train one model for both conditional & unconditional

(Ho & Salimans, 2022)

Classifier-Free Guidance (CFG) achieves strong guidance *without* needing any external classifier.

Diffusion Models in 30 Seconds

Forward Process $q(x_t|x_0)$

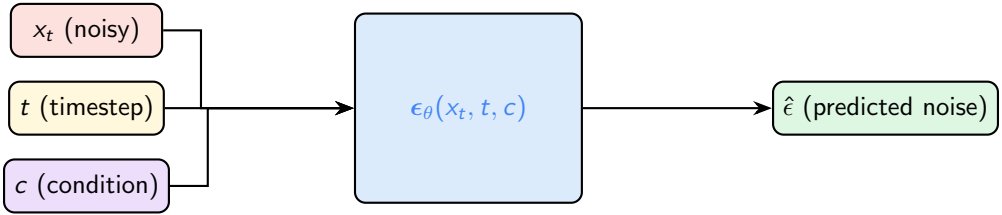


Key idea: The model $\epsilon_\theta(x_t, t)$ predicts the noise ϵ that was added to get x_t, then we subtract it to denoise step-by-step.

Conditional Diffusion Models

To generate images conditioned on a label/text c, we train:

$\epsilon_\theta(x_t, t, c) \quad \longleftarrow$ "predict noise, given noisy image x_t, timestep t, and condition c"



Problem: This alone often doesn't produce images that strongly match the condition c. We need a way to **amplify** the effect of the condition.

Classifier Guidance: The Idea

Dhariwal & Nichol (2021) proposed using a pretrained classifier $p_\phi(c|x_t)$ to guide the diffusion process.

The score function of a conditional model can be decomposed via Bayes' rule:

$$\nabla_{x_t} \log p(x_t|c) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(c|x_t)$$

$\nabla_{x_t} \log p(x_t)$ Unconditional score	$+$	$\nabla_{x_t} \log p(c\|x_t)$ Classifier gradient	$=$	$\nabla_{x_t} \log p(x_t\|c)$ Conditional score

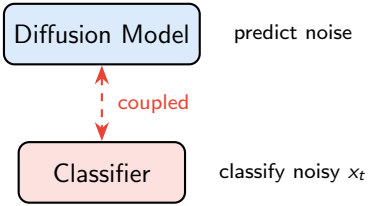
Add a guidance scale s to strengthen the classifier signal:

$$\tilde{\nabla}_{x_t} = \nabla_{x_t} \log p(x_t) + s \cdot \nabla_{x_t} \log p(c|x_t)$$

Classifier Guidance: The Drawbacks

Downsides:

1. Need to train a **separate classifier** on noisy images x_t at every noise level
2. The classifier must handle **all noise levels** $t = 1, \ldots, T$
3. Classifier gradients can be **adversarial** — pushing images toward classifier-fooling artifacts
4. More complex pipeline: two models to maintain



Two separate models!

Can we get the same guidance effect with just *one* model?

The Key Insight

Core Observation

From Bayes' rule: $\nabla_{x_t} \log p(c|x_t) = \nabla_{x_t} \log p(x_t|c) - \nabla_{x_t} \log p(x_t)$

Substitute back into the guided score:

$$\tilde{\nabla}_{x_t} = \nabla_{x_t} \log p(x_t) + s \cdot \left[\nabla_{x_t} \log p(x_t|c) - \nabla_{x_t} \log p(x_t)\right]$$

$$= \underbrace{(1-s) \cdot \nabla_{x_t} \log p(x_t)}_{\text{unconditional}} + \underbrace{s \cdot \nabla_{x_t} \log p(x_t|c)}_{\text{conditional}}$$

No Classifier Needed!

We only need:

1. An **unconditional** score $\nabla_{x_t} \log p(x_t) \quad \Leftrightarrow \quad \epsilon_\theta(x_t, t)$
2. A **conditional** score $\nabla_{x_t} \log p(x_t|c) \quad \Leftrightarrow \quad \epsilon_\theta(x_t, t, c)$

Both can come from the **same** diffusion model!

The CFG Formula

In terms of the noise prediction network ϵ_θ:

$$\boxed{\tilde{\epsilon}_\theta(x_t, t, c) = \underbrace{\epsilon_\theta(x_t, t, \varnothing)}_{\text{unconditional}} + w \cdot \Big(\underbrace{\epsilon_\theta(x_t, t, c)}_{\text{conditional}} - \underbrace{\epsilon_\theta(x_t, t, \varnothing)}_{\text{unconditional}} \Big)}$$

where $w \geq 0$ is the **guidance scale** and $\varnothing$ denotes "no condition" (null/empty).

$\epsilon_\theta(x_t, t, \varnothing)$
"Where is the image heading in general?"

→

$\epsilon_\theta(x_t, t, c)$
"Where should it go to match condition c?"

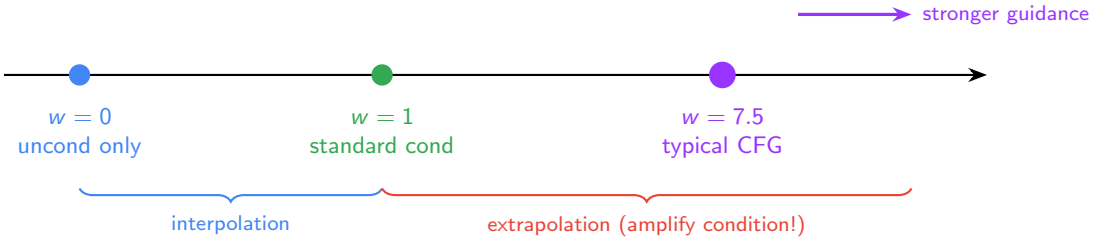
→

$\Delta = \text{cond} - \text{uncond}$
"The direction that makes it more like c"

Equivalent Form: Linear Interpolation / Extrapolation

Rearranging the CFG formula:

$$\tilde{\epsilon} = (1 - w) \cdot \epsilon_\theta(x_t, t, \varnothing) + w \cdot \epsilon_\theta(x_t, t, c)$$

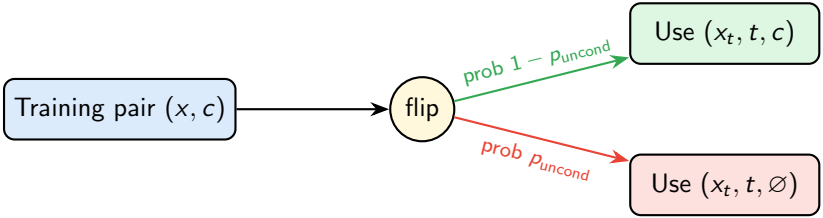


- $w = 0$: Ignore the condition entirely (unconditional generation)
- $w = 1$: Standard conditional generation (no guidance amplification)
- $w > 1$: **Extrapolate** away from unconditional $\Rightarrow$ **amplify** the condition
- Typical values: $w \in [3, 15]$ depending on the task

Training Strategy: Random Dropout of Condition

Key trick: Train a *single* model that can do both conditional and unconditional prediction.

During training, randomly replace the condition c with a null token $\varnothing$ with probability p_{uncond} (typically 10–20%):



Same model ϵ_θ handles both!

Loss (same as standard diffusion):

$$\mathcal{L} = \mathbb{E}_{x_0,\epsilon,t,c'}\left[\left\|\epsilon - \epsilon_\theta(x_t, t, c')\right\|^2\right], \quad c' = \begin{cases} c & \text{w.p. } 1 - p_{\text{uncond}} \\ \varnothing & \text{w.p. } p_{\text{uncond}} \end{cases}$$

What is the Null Condition $\varnothing$?

Different implementations handle $\varnothing$ differently:

Model	**Condition type**	$\varnothing$ **implementation**
Class-conditional	Class label $c \in \{0, \ldots, K\}$	Extra "null class" ID
Text-to-image	Text embedding	Empty string "" embedding
DALL-E 2	CLIP embedding	Zero vector
Stable Diffusion	Text tokens	Embedding of "" (empty)

The key point: $\varnothing$ tells the model "I have no condition — just generate anything plausible."

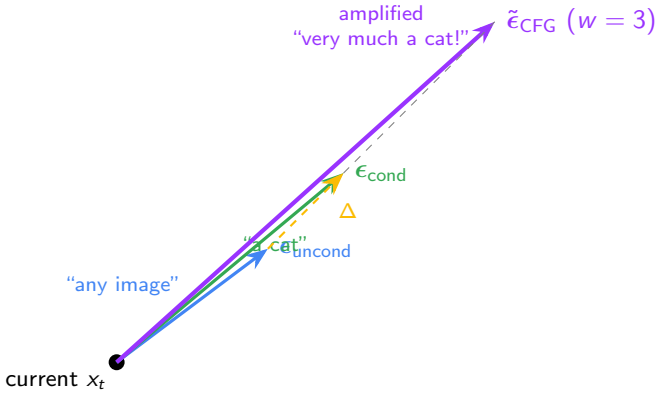
Inference: Step by Step

At each denoising step $t = T, T-1, \ldots, 1$:

1. $\boldsymbol{\epsilon}_{\text{uncond}} = \boldsymbol{\epsilon}_\theta(x_t, t, \varnothing)$ Unconditional prediction
2. $\boldsymbol{\epsilon}_{\text{cond}} = \boldsymbol{\epsilon}_\theta(x_t, t, c)$ Conditional prediction
3. $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\text{uncond}} + w \cdot (\boldsymbol{\epsilon}_{\text{cond}} - \boldsymbol{\epsilon}_{\text{uncond}})$
4. Use $\tilde{\epsilon}$ to compute x_{t-1} (DDPM/DDIM step)

Note: Each step requires **two** forward passes (one unconditional, one conditional). This doubles the compute cost compared to unguided sampling.

Visual Intuition: What CFG Does



Intuition: CFG takes the "direction toward condition c" (the difference between conditional and unconditional predictions) and **amplifies** it by factor w.

Effect of Guidance Scale w

Guidance Scale: Practical Examples

w	**Behavior**	**Analogy**
0	Pure unconditional generation	"Draw whatever you want"
1	Standard conditional (no boost)	"Draw a cat"
3	Mild guidance	"Draw a cat, please try harder"
7.5	**Typical for Stable Diffusion**	"Draw a *really good* cat"
15	Very strong guidance	"ONLY a cat, nothing else!"
50+	Over-saturated, artifacts	Distorted, unnatural

Trade-off:

- Higher $w \Rightarrow$ better alignment with condition, but less diversity and potential artifacts
- Lower $w \Rightarrow$ more diverse, but weaker condition adherence

Formal Derivation

Starting point: We want to sample from $p(x_t|c)$ but amplified.

The score of $p(x_t|c)$ using Bayes' rule:

$$\nabla_{x_t} \log p(x_t|c) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(c|x_t)$$

With guidance scale s:

$$\begin{aligned}\tilde{\nabla}_{x_t} &= \nabla_{x_t} \log p(x_t) + s \cdot \nabla_{x_t} \log p(c|x_t) \\ &= \nabla_{x_t} \log p(x_t) + s \cdot \left[\nabla_{x_t} \log p(x_t|c) - \nabla_{x_t} \log p(x_t)\right] \\ &= (1-s)\, \nabla_{x_t} \log p(x_t) + s\, \nabla_{x_t} \log p(x_t|c)\end{aligned}$$

Since $\epsilon_\theta \approx -\sigma_t \nabla_{x_t} \log p(x_t)$, we translate to ϵ-space (with $w = s$):

$$\boxed{\tilde{\epsilon} = \epsilon_\theta(x_t, t, \varnothing) + w \cdot \left[\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)\right]}$$

This is the **Classifier-Free Guidance** formula. No classifier anywhere!

Connection to Implicit Classifier

CFG *implicitly* uses the diffusion model itself as a classifier:

$$\nabla_{x_t} \log p_{\text{implicit}}(c|x_t) \;\propto\; \epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)$$

Classifier Guidance
Uses explicit $p_\phi(c|x_t)$
(separate network)

mathematically equivalent form

Classifier-Free Guidance
Uses implicit classifier
(same network, two passes)

The diffusion model "knows" what c looks like vs. what a generic image looks like. The *difference* between the two predictions acts as an implicit classifier gradient.

Cost: Two Forward Passes

Main downside: CFG requires **2 network evaluations per step**.

Without CFG:	1 forward pass	$\epsilon_\theta(x_t, t, c)$
With CFG:	Pass 1 (uncond)	$\epsilon_\theta(x_t, t, \varnothing)$
	Pass 2 (cond)	$\epsilon_\theta(x_t, t, c)$

Optimization: Batch both passes together — send $[x_t, x_t]$ with conditions $[\varnothing, c]$ as a batch of size 2. This uses GPU parallelism efficiently.

Distillation: Recent work (e.g., Meng et al., 2023) trains a single model to directly predict the CFG-combined output, eliminating the double cost.

CFG in Popular Models

Model	**Uses CFG?**	**Typical w**
DALL-E 2 (OpenAI)	Yes	$\sim$ 2–4
Imagen (Google)	Yes	$\sim$ 7–10
Stable Diffusion	Yes	$\sim$ 7.5 (default)
Midjourney	Yes (variant)	Not disclosed
SDXL	Yes	$\sim$ 5–9
PixArt-α	Yes	$\sim$ 4.5

CFG is used in **virtually all** modern text-to-image diffusion models.
It is one of the most impactful techniques in generative AI.

Summary: CFG in One Slide

Training
Randomly drop condition (p_{uncond})

One model learns both conditional & unconditional

Inference
$\tilde{\epsilon} = \epsilon_{\varnothing} + w(\epsilon_c - \epsilon_{\varnothing})$

$w > 1$ extrapolates toward the condition

Effect
Amplify condition, improve quality

Better FID, better text-image alignment

Cost
$2\times$ forward passes per step

Batch trick or distillation to mitigate

Key Takeaways

1. **Problem:** Conditional diffusion models need "guidance" to produce high-quality, condition-aligned outputs.

2. **Old solution (Classifier Guidance):** Use a separate classifier $p(c|x_t)$ to steer generation. Requires extra model, noisy-image training.

3. **CFG solution:** Train one model with random condition dropout. At inference, **extrapolate** from unconditional toward conditional prediction:

$$\tilde{\epsilon} = \epsilon_{\varnothing} + w\left(\epsilon_c - \epsilon_{\varnothing}\right)$$

4. **Result:** Simple, effective, no extra models. Used in DALL-E 2, Stable Diffusion, Imagen, and almost all modern text-to-image systems.

References

- **[Ho & Salimans, 2022]** J. Ho and T. Salimans. "Classifier-Free Diffusion Guidance." *NeurIPS Workshop on DGMs*, 2022.
- **[Dhariwal & Nichol, 2021]** P. Dhariwal and A. Nichol. "Diffusion Models Beat GANs on Image Synthesis." *NeurIPS*, 2021.
- **[Ho et al., 2020]** J. Ho, A. Jain, and P. Abbeel. "Denoising Diffusion Probabilistic Models." *NeurIPS*, 2020.
- **[Meng et al., 2023]** C. Meng et al. "Distillation of Guided Diffusion Models." *CVPR*, 2023.